

BHU-NETRA: Satellite Ground Segment & Mission Control Platform

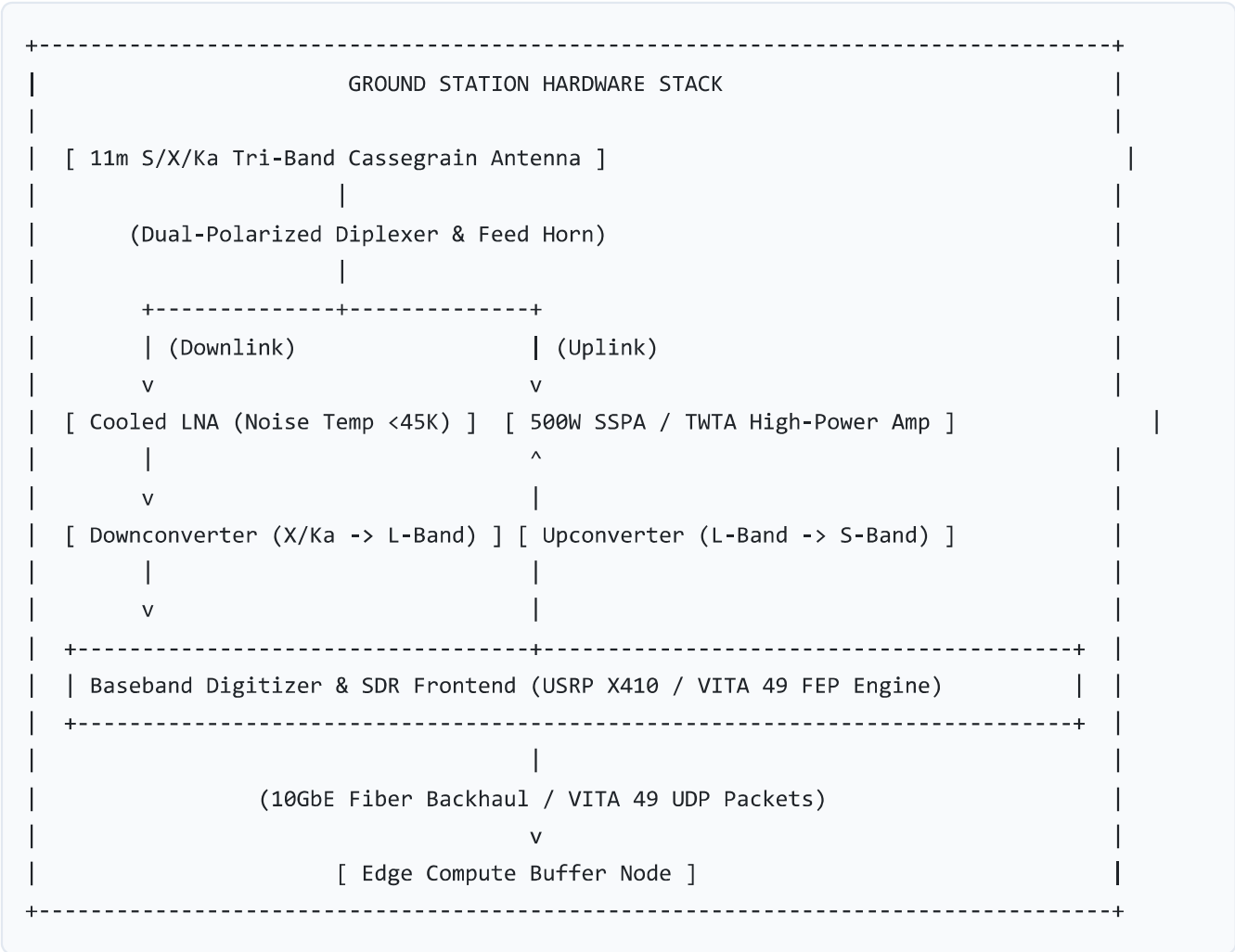
Document 05: Hardware Specifications & Ground Infrastructure Engineering

1. Overview & Operational Environment

BHU-NETRA operates across a hybrid physical infrastructure encompassing:

- Primary Mission Operations Complex (MOX):** Centralized, air-gapped tier-IV data center housing core C2, telemetry, and HPC geo-processing clusters.
- Domestic Ground Station Network:** Deep-space and EO ground stations (e.g., Shadnagar, Hyderabad; Hassan, Karnataka; Port Blair, A&N).
- International / Polar Ground Stations:** Svalbard, Norway; Antarctica (Bharati Station); and polar downlink facilities.
- Cloud Dissemination DMZ:** Hosted on government-accredited cloud infrastructure.

2. Ground Station RF, Antenna Pedestal & Baseband Hardware



2.1 Antenna Systems & RF Frontends

- **Antenna Reflector:** 11-meter & 7.5-meter Cassegrain Tri-Band (S-Band, X-Band, Ka-Band) parabolic tracking dish with monopulse auto-tracking feed.
- **Tracking Servo Controller:** Antenna Control Unit (ACU) with dual-axis brushless AC servo motors providing 15 deg/sec azimuth and 10 deg/sec elevation slew rates.
- **Low Noise Amplifiers (LNA):** Cryogenically/thermo-electrically cooled X-band LNA (Noise Temperature < 45 K; G/T ratio > 32 dB/K at 5 deg elevation).
- **High Power Amplifiers (HPA):** Dual 500W Solid-State Power Amplifiers (SSPA) with automatic switchover for S-Band command uplink.

2.2 Baseband & Software Defined Radio (SDR) Hardware

- **Digitizer / Front-End Processor (FEP):** USRP X410 Quad-Channel SDR with Dual 100GbE interfaces and Xilinx Zynq UltraScale+ RFSoc.

- **VITA 49 Engine:** Hardware packetizer digitizing RF IF (Intermediate Frequency) signals directly into VITA 49 VRT streams over 10GbE optical links.
- **Legacy FEP Converter:** PCI-Express / CPCI telemetry demodulator cards handling legacy PCM/FM, BPSK, and QPSK modulation formats for older ground station assets.

3. On-Premise High-Performance Compute (HPC) & Server Infrastructure

3.1 Mission Operations & C2 Core Cluster

- **Master Control Server Nodes (x6 HA Nodes):**
 - *Processor:* Dual AMD EPYC 9654 (96 Cores, 192 Threads per CPU, 2.4 GHz base).
 - *RAM:* 1.5 TB ECC DDR5-4800 MHz.
 - *Network:* Dual-port 100GbE Mellanox ConnectX-6 SmartNICs with RoCEv2 support.
 - *OS:* Hardened Rocky Linux 9 (Air-gapped kernel with SELinux Enforcing).

3.2 Real-time Geo-Processing GPU Cluster (Level-0 to Level-3 Ingest)

- **HPC Compute Nodes (x16 Nodes):**
 - *Processor:* Dual Intel Xeon Platinum 8480+ (56 Cores per CPU).
 - *GPUs:* 4x NVIDIA H100 80GB SXM5 GPUs per node (Total 64 H100 GPUs across cluster) connected via NVLink for parallel orthorectification, SAR phase processing, and radiometric calibration.
 - *RAM:* 1 TB ECC DDR5.
 - *Storage:* Local 15.36 TB NVMe U.3 PCIe 5.0 Scratch Disk per node.

4. Multi-Tier Storage Subsystems Architecture

Storage Tier	Technology Stack	Capacity	Read/Write Throughput	Purpose
Tier 0 (Hot NVMe Array)	Pure Storage FlashArray//XL or Custom Ceph NVMe Cluster	200 Terabytes	100 Gigabits/sec (Direct DMA)	Real-time pass ingestion, raw bitstream demuxing
Tier 1 (Warm SAN/NAS)	MinIO Enterprise / NetApp StorageGRID (100GbE Backhaul)	5 Petabytes	40 Gigabits/sec	90-day active imagery COG store & PostGIS spatial catalog
Tier 2 (Cold Tape Archive)	IBM TS4500 LTO-9 Enterprise Tape Library (Dual Robotics)	18 Petabytes (Expandable)	400 Megabytes/sec per drive	Permanent air-gapped national satellite payload archive

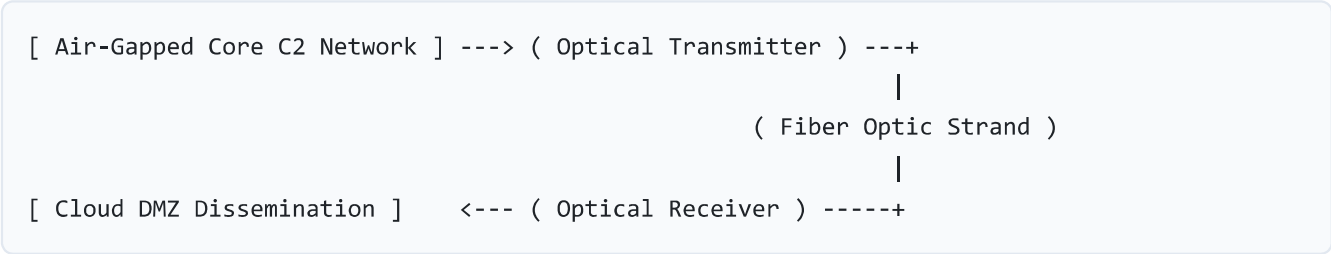
5. Cryptographic Security Enclave & Network Hardware

5.1 Command Enclave Cryptographic Hardware

- **Hardware Security Modules (HSM):** Dual Thales Luna HSM 7 (FIPS 140-3 Level 3 certified) installed in tamper-evident server racks.
- **Smart Card / YubiKey Authenticators:** YubiKey 5 Series FIPS tokens required for physical insertion by Mission Director and FDO during telecommand sign-off.

5.2 Unidirectional Optical Data Diodes

- **Hardware Diodes:** Advenica SecuriCDS / OWL Cyber Defense 10Gbps Hardware Data Diodes.
- **Function:** Physically enforces ONE-WAY data flow from air-gapped Mission Control to DMZ cloud portal, ensuring zero inbound TCP connection can reach C2 networks.



5.3 Power & Network Redundancy

- **Core Network Switches:** Spine-Leaf Architecture with Arista 7060X4 32x 100GbE Switches.
- **Firewalls:** High-Availability Pair of Palo Alto PA-5450 Next-Gen Firewalls.
- **Uninterruptible Power Supply (UPS):** 300 kVA Dual-Redundant Online UPS with Lithium-ion Battery Banks providing 45 minutes of full-load backup, backed by twin 750 kVA Diesel Generators with automatic transfer switches (ATS).